

Taxonomy of Multimodal Time Series Models

Multimodal Time Series Paradigm (Data, Fusion, Role, Task, Domain)

1. Modality Pairing

- TS + Text (Reports, News, Prompts)
- TS + Vision (Plots, Spectrograms, MAE)
- TS + Tactile Skin Arrays (RDP, GelNeuro)
- TS + Neuromorphic DVS (REACT, Colibri)
 - TS + Cryptographic Proofs (zkSTAR)
- TS + Planetary Teleconnections (STM3)

2. Fusion Architecture

- Patch Reprogramming (Time-LLM, OFA)
 - Visuotactile Diffusion Policy (RDP)
- Symplectic Flow Matching (CoSynFlow)
- Causal DAG Attention (CCPF, MATERO)
- State-Space zk-SNARK Prover (zkSTAR)
- Continuous Neural ODE / CDE (SOTER)

3. Non-TS Role

- Auxiliary Context / Condition
- High-Frequency Reactive Force Prior
- Symplectic Invariant Manifold (HamNO)
 - zk-SNARK Compliance Certificate
 - Pearl Causal Parent Mask (CCPF)
- On-Chip Neuromorphic Spike Trigger

4. Downstream Tasks

- Contact-Rich Dexterous Manipulation
 - Multi-Phase Turbulence Telemetry
- Verifiable Power Substation Defense
- Fab Root-Cause Anomaly Attribution
- Multi-Decadal S2S Climate Prediction
- Sub-50mW Neuromorphic Flight Control

Application Domains: Robotics & Manipulation • Healthcare & EHR • Meteorology & Climate • Finance & Markets • Energy & Smart Grids • Industrial IoT & 6G